19.23 Chain rule

As we know from §10.12 that formula (V), i.e.

$$\frac{dz}{dx} = \frac{dz}{dy}\cdot\frac{dy}{dx}$$

is the formula for differentiating a *composite function.* To justify this formula in the vector case, we therefore need to establish the following result. (See theorem 10.13.)

19.24 Proposition Suppose that g is differentiable at ξ and f is differentiable at $\eta = g(\xi)$. Then $f \circ g$ is differentiable at ξ and

$$(f \circ g)'(\xi) = f'(\eta)g'(\xi).$$

Proof We are given that there exist matrices $f'(\eta)$ and $g'(\xi)$ for which

$$f(y) - f(\eta) - f'(\eta)(y - \eta) = o(y - \eta) \quad (y \to \eta)$$

$$g(x) - g(\xi) - g'(\xi)(x - \xi) = o(x - \xi) \quad (x \to \xi).$$

Writing $y = g(x)$ and $\eta = g(\xi)$ in the first equation, we obtain that

$$f(g(x)) - f(g(\xi)) - f'(\eta)(g(x) - g(\xi)) = o(g(x) - g(\xi))$$

and hence

$$f(g(x)) - f(g(\xi)) - f'(\eta)\{g'(\xi)(x - \xi) + o(x - \xi)\}$$
$$= o\{g'(\xi)(x - \xi) + o(x - \xi)\}.$$

It follows that

$$f(g(x)) - f(g(\xi)) - f'(\eta)g'(\xi)(x - \xi) = o(x - \xi) \quad (x \to \xi)$$

and so $f \circ g$ is differentiable at ξ with derivative $f'(\eta)g'(\xi)$. As in theorem 10.13, the final step is not entirely straightforward but we prefer not to hold up the discussion by presenting a detailed proof at this stage.

Written in full, formula (V) takes the form

$$\begin{pmatrix} \frac{\partial z_1}{\partial x_1} & \frac{\partial z_1}{\partial x_2} & \cdots & \frac{\partial z_1}{\partial x_n} \\ \frac{\partial z_2}{\partial x_1} & \frac{\partial z_2}{\partial x_2} & \cdots & \frac{\partial z_2}{\partial x_n} \\ \vdots & \vdots & & \\ \frac{\partial z_m}{\partial x_1} & \frac{\partial z_m}{\partial x_2} & \cdots & \frac{\partial z_m}{\partial x_n} \end{pmatrix} = \begin{pmatrix} \frac{\partial z_1}{\partial y_1} & \cdots & \frac{\partial z_1}{\partial y_p} \\ \frac{\partial z_2}{\partial y_1} & \cdots & \frac{\partial z_2}{\partial y_p} \\ \vdots & & \\ \frac{\partial z_m}{\partial y_1} & \cdots & \frac{\partial z_m}{\partial y_p} \end{pmatrix} \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \cdots & \frac{\partial y_1}{\partial x_n} \\ \vdots & & & \\ \frac{\partial y_p}{\partial x_1} & \frac{\partial y_p}{\partial x_2} & \cdots & \frac{\partial y_p}{\partial x_n} \end{pmatrix}$$

from which we deduce that

$$\frac{\partial z_i}{\partial x_j} = \frac{\partial z_i}{\partial y_1}\frac{\partial y_1}{\partial x_j} + \frac{\partial z_i}{\partial y_2}\frac{\partial y_2}{\partial x_j} + \ldots + \frac{\partial z_i}{\partial y_p}\frac{\partial y_p}{\partial x_j}.$$

This formula is called the *chain rule* and is often very useful in calculating partial derivatives.

19.25 Examples

(i) Let f: $\mathbb{R}^2 \to \mathbb{R}^1$ and g: $\mathbb{R}^1 \to \mathbb{R}^2$ be defined by

$$z = f(y_1, y_2) = y_1^2 y_2 + y_2^3 y_1 ; y = g(x) = \begin{pmatrix} x^2 \\ x^3 \end{pmatrix}.$$

From formula (V)

$$\frac{dz}{dx} = \frac{dz}{dy}\cdot\frac{dy}{dx} = (2y_1y_2 + y_2^3, y_1^2 + 3y_2^2y_1)\begin{pmatrix} 2x \\ 3x^2 \end{pmatrix}.$$

This has to be calculated when $y = g(x)$, i.e. when $y_1 = x^2$ and $y_2 = x^3$. Thus

$$\frac{dz}{dx} = (2x^5 + x^9, x^4 + 3x^8)\begin{pmatrix} x^2 \\ x^3 \end{pmatrix} = 7x^6 + 11x^{10}.$$

The same result can be easily obtained directly in this case because $f \circ g$: $\mathbb{R}^1 \to \mathbb{R}^1$ is given by the simple formula $f \circ g(x) = (x^2)^2x^3 + (x^3)^3x = x^7 + x^{11}$.

(ii) Suppose that $w = xy^2z$ where

$$x = r + 2s + 3t$$

$$y = 2r + 3s + t$$

$$z = 3r + s + t.$$

We can calculate the partial derivative of w with respect to r, s or t using the chain rule. For example

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial s}.$$

It is as well to appreciate that every symbol in this formula is used ambiguously. This can naturally lead to confusion unless one is quite clear about what everything means. It is sometimes helpful to rewrite the equation in the form

$$\left(\frac{\partial w}{\partial s}\right)_{r,t} = \left(\frac{\partial w}{\partial x}\right)_{y,z}\left(\frac{\partial x}{\partial s}\right)_{r,t} + \left(\frac{\partial w}{\partial y}\right)_{x,z}\left(\frac{\partial y}{\partial s}\right)_{r,t} + \left(\frac{\partial w}{\partial z}\right)_{x,y}\left(\frac{\partial z}{\partial s}\right)_{r,t}$$

to make it clear what is to be held constant in each differentiation. (See § 19.5.) Thus

$$\frac{\partial w}{\partial s} = y^2z^32 + 2xyz^33 + 3xy^2z^2 = 2(2r + 3s + t)^2(3r + s + t)^3$$
$$+ 6(r + 2s + 3t)(2r + 3s + t)(3r + s + t)^3$$
$$+ 3(r + 2s + 3t)(2r + 3s + t)^2(3r + s + t)^2.$$

We now turn to formula (VI), i.e.

$$\frac{dy}{dx} = \left(\frac{dx}{dy}\right)^{-1}.$$

As we know from § 12.9, this formula concerns the derivative of an *inverse function.* Assuming that f has an inverse function f^{-1}, then

$$f \circ f^{-1}(x) = x$$

(exercise 7.16(5)). If f is differentiable at ξ and f^{-1} is differentiable at $\eta = f(\xi)$, then the formula (V) yields that

$$f'(\xi)(f^{-1})'(\eta) = I$$

from which formula (VI) follows.

When considering the real case we also gave conditions under which a suitable differentiable inverse exists. It follows from theorem 12.10 that it is sufficient in the real case if f' exists and is positive on some open interval containing ξ. The same is, of course, true if the word positive is replaced by the word negative. In the particular case when f' is continuous at ξ, it is therefore enough if $f'(\xi) \neq 0$.

A similar result holds in the vector case. Note that $f'(\xi) \neq 0$ in the real case corresponds to the requirement that the affine function tangent to f at ξ has an inverse function. The condition in the vector case which corresponds to this requirement is that the $n \times n$ matrix $f'(\xi)$ be *non-singular*, i.e.

$$\det f'(\xi) \neq 0.$$

We call the determinant $\det f'(\xi)$ the *Jacobian* of f at ξ. It is an important result that a suitable differentiable inverse of f exists provided that all the first order partial derivatives of f are continuous at ξ and $\det f'(\xi) \neq 0$. However, we shall neither explain the meaning of the word 'suitable' in this context nor offer a proof of the result. A proof would require apparatus that has not been developed in this book.

19.26 Exercise

(1) Explain why all functions f: $\mathbb{R}^n \to \mathbb{R}^m$ whose co-ordinates are all polynomials are differentiable at every point. What is the appropriate result when the co-ordinates are rational functions?

(2) Explain why each of the following functions is differentiable at the given point and compute its derivative there.

(i) $f\colon \mathbb{R}^2 \to \mathbb{R}$ at $(1, 2)$ where $f(x_1, x_2) = x_1^2 + 2x_1x_2 + x_2^3$.

(ii) $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ at $(\pi, 1)$ where

$$f(x_1, x_2) = \begin{pmatrix} \cos(x_1x_2) \\ \sin(x_1x_2) \end{pmatrix}.$$

(iii) $f\colon \mathbb{R}^2 \to \mathbb{R}^3$ at $(1, 1)$ where

$$f(x_1, x_2) = \begin{pmatrix} x_1 \\ x_1 + x_2 \\ x_1x_2 \end{pmatrix}.$$

(iv) $f\colon \mathbb{R}^3 \to \mathbb{R}^2$ at $(1, 2, 3)$ where

$$f(x_1, x_2, x_3) = \begin{pmatrix} x_1x_2^2 \\ x_2^2x_3^3 \end{pmatrix}.$$

(v) $f\colon \mathbb{R} \to \mathbb{R}^3$ at 0 where

$$f(x) = \begin{pmatrix} 1 \\ x \\ x^2 \end{pmatrix}$$

For each of these functions, write down the equations for the appropriate tangent flat.

(3) What is the slope (or rate of increase) of the function f of question 2(i) at the point $(1, 2)$ in the direction $(1, -1)$? What is the direction at $(1, 2)$ in which the slope is greatest? What is the value of the slope in this direction? Write down a vector which is normal to the contour $x_1^2 + 2x_1x_2 + x_2^3 = 13$ at $(1, 2)$. Also write down the equation of the tangent line at this point.

(4) Explain the circumstances under which formulae (I), (III) and (IV)′ of §19.21 make sense when L and M are matrices and x, y, z and c are column vectors. Prove that the formulae are valid when they make sense.

(5) Suppose that $f\colon \mathbb{R} \to \mathbb{R}$ is differentiable everywhere and that the function $\phi\colon \mathbb{R}^2 \to \mathbb{R}$ is defined by $\phi(x_1, x_2) = f(x_1x_2)$. If $y = \phi(x_1, x_2)$, prove that

$$x_1 \frac{\partial y}{\partial x_1} = x_2 \frac{\partial y}{\partial x_2}$$

for all $(x_1, x_2) \in \mathbb{R}^2$ for which $f'(x_1x_2) \neq 0$.